

TrueMatch

Platform & Companion App Briefing

The AI-Native, Capability-First Recruitment & Assessment System

Features, Functions & Capabilities — by Administrator, Recruiter & Candidate

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1. Executive Summary

TrueMatch is an **AI-native recruitment and assessment platform** built on a single conviction: hiring systems reward **credentials and keywords**, while what actually predicts performance is **demonstrated capability**. Conventional Applicant Tracking Systems (ATS) match a résumé's words against a job description's words. TrueMatch instead **reasons** about what a candidate can actually do — and surfaces the gap between the two.

The platform is delivered as a **monorepo** spanning three layers: a FastAPI reasoning backend, a Next.js web application, and a native SwiftUI iOS companion app. It serves three roles — **Administrator, Recruiter, and Candidate** — each with a tailored experience over a shared, governed reasoning core.

Its signature output is a **three-signal score**: a traditional keyword/ATS score, a semantic-match score, and a capability score — and the **delta** between them, which is the product. A large positive delta flags a **hidden-gem candidate** the conventional ATS would have filtered out.

Every AI verdict passes mandatory **governance gates**, is recorded with a **reproducibility manifest** for audit, and runs on a **redundant dual-LLM** backbone (primary plus automatic failover) so the service stays available. The platform also includes a complete **commercial layer** — payments, a finite "Founding-100" launch programme, a two-sided referral loop, and shareable anonymised results.

2. Background & The Problem

Recruitment funnels are overwhelmed. To cope, employers lean on automated keyword filters that reject the majority of applicants before a human reads them. This produces two well-documented failures:

- **False negatives (hidden gems):** capable candidates are filtered out because their résumé phrases real experience in different words than the job description (e.g. "Docker Swarm" when the posting says "Kubernetes").
- **False positives (keyword stuffers):** candidates who echo the posting's vocabulary rank highly regardless of whether they can do the work.

TrueMatch was created to correct this by making capability **legible** — to recruiters (so they see the candidates the ATS hid) and, on the candidate side, by translating genuine, evidenced capability into the language the gatekeeping systems understand. It is built compliance-first for regulated markets (Singapore PDPA, EU AI Act, NYC Local Law 144).

3. The Core Thesis: Capability Before Credential

TrueMatch evaluates every résumé–role pairing along three independent signals and reports the difference between them:

Signal	What it measures	How it is computed
Traditional / Keyword	How a conventional ATS would rank the résumé against the JD	Deterministic keyword & term-frequency overlap (reproducible)
Semantic	Conceptual fit beyond exact words	Deterministic text-embedding similarity of résumé and JD spans
Capability	What the candidate has actually demonstrated	AI reasoning over evidence, trajectory and verified signals
Delta (the product)	The gap between keyword rank and real capability	Capability minus Traditional — large positive = hidden gem

The deterministic signals are **byte-for-byte reproducible** run-to-run; the capability signal is AI-reasoned and governed. Presenting all three together is the platform's defining differentiator — independent market analysis of 33 AI recruitment platforms found none that expose this gap.

4. System Architecture

TrueMatch is a single monorepo with three deployable layers over one reasoning core:

Layer	Technology	Responsibility
Backend	FastAPI · SQLAlchemy (async) · Celery · PostgreSQL · Redis	Reasoning engines, governance, persistence, APIs, async pipelines
Web app	Next.js 14 (App Router) · Tailwind · Recharts	Role dashboards for admin, recruiter, candidate (browser)
iOS app	SwiftUI (iOS 17+) · MVVM	Native companion client (role-aware tabs)
AI core	Claude (primary) + MiniMax (failover) · on-device embeddings	Capability reasoning, scoring and translation engines

A **BFF proxy** in the web layer injects the authenticated user's token server-side, so the browser never holds backend credentials. Heavy, multi-step AI pipelines run asynchronously on Celery workers and are polled for completion. All external integrations (payments, transcription, calendar, ATS connectors, e-mail, push) are **gated**: each is a safe no-op until configured, so nothing breaks when a key is absent.

5. The AI Reasoning Core — Eight Capability Pillars

The capability score is not a single prompt; it is the product of eight cooperating reasoning pillars:

- 1 **Three-Signal Architecture** — keyword, semantic and capability scores plus the delta.
- 2 **Credential Substitution** — identifies where a résumé's alternate evidence satisfies a credential the role demands, each rated HIGH / MEDIUM / WEAK with cited evidence.

- 3 **Counter-Recommendation** — proactively surfaces candidates whose capability exceeds the role despite a low keyword match — the hidden-gem signal.
- 4 **JD Interrogation** — scores the job description's own quality 0–100, flags inflated or impossible requirements, and suggests fixes.
- 5 **JD Evolution Tracking** — versions every job description and detects drift over time (experience creep, scope expansion, quality decline).
- 6 **External Evidence Verification** — turns self-asserted claims into provenanced evidence against authoritative open registries (ORCID, DataCite, Crossref, Zenodo, GitHub), discovers a candidate's published work by identity, and corroborates non-API claims (e.g. patents) from official supporting documents — using a three-tier trust model (verified / corroborated / claim) so capability is never inflated by self-supplied material. Detailed in Appendix B.
- 7 **Semantic Matching** — a deterministic embedding signal that differentiates conceptual fit independently of the AI verdict.
- 8 **Autonomous Agentic Operation** — ingestion via folder drop / e-mail / API, scheduled processing, auto-approve / escalate routing, and a durable cross-session plan runner.

A newer, candidate-facing pillar — **Capability Translation** — completes the thesis: it re-expresses a candidate's **evidenced** capability into ATS-legible language (never fabricating), then re-scores the rewrite with the very same keyword and semantic engines to **prove the measured lift**. It is the candidate-side mirror of credential substitution.

6. Governance, Provenance & Compliance

Every assessment must pass mandatory, non-bypassable **governance gates** before it is surfaced:

- **Coherence** — do the component scores, evidence and narrative agree with one another?
- **Consistency** — is the verdict stable and internally non-contradictory?
- **Fidelity** — is every claim grounded in the source résumé (no fabrication)?
- **Bias scan** — flags reasoning that leans on protected or proxy attributes.

Gate thresholds live **server-side and encrypted**; the client displays gate outcomes only. A failed gate routes the assessment to human review rather than auto-deciding. Each run also emits a **reproducibility manifest** (input hashes, model and engine versions, method identifiers) recorded to the audit trail, giving legal defensibility under the EU AI Act and NYC LL144. PDPA/GDPR data-subject rights are built in: candidates can export their data and request erasure.

7. Monetisation & Growth Layer

TrueMatch includes a complete commercial layer, all gated and webhook-driven, with card data never touching the platform's servers (PCI SAQ A via hosted checkout):

- **Payments & catalogue** — pay-per-assessment, bundles, CV/JD reports, team and subscription plans; promotions and Singapore PayNow supported via the processor.

- **Credits & entitlements** — an append-only credit ledger and plan entitlements (founding 12-month, subscription, unlimited).
- **Founding-100 programme** — a finite launch inventory (60 candidate / 30 enterprise / 10 partner) with a live remaining-spots counter and a sold-out gate.
- **Two-sided referrals** — both referrer and referee earn a free credit, with anti-abuse (a user can be referred only once).
- **Shareable anonymised results** — a candidate can share a public snapshot of their delta and counter-recommendation with no personal data, driving the referral loop.
- **Admin fulfilment queue & refunds** — an operator console to advance order fulfilment and process refunds.

8. Capabilities by Role

8.1 Candidate

Candidates use TrueMatch to understand and improve how they are seen by hiring systems. Available on both web and the iOS companion app.

Capability	What it does
Self-assessment	Upload a résumé, paste a target JD, and receive the three-signal score with the capability narrative and delta.
Capability Translation	A grounded, ATS-legible rewrite with measured before→after keyword and semantic lift, per-bullet evidence grounding, and an honest list of genuine gaps not added.
CV Analysis	Career-level gap analysis vs a target role/seniority: strengths, missing capabilities, matching roles, growth opportunities, trajectory.
AI Assistant	A conversational agent for guidance, with streaming responses.
Jobs	Browse and match against open positions.
Assessment history	Review past assessments and results.
Refer & earn / Pricing	Share a referral link for mutual free credits; purchase credits or plans.
Shareable result	Publish an anonymised result snapshot (no PII) to showcase the capability–keyword gap.
Data rights	Export personal data or request erasure (PDPA/GDPR).

8.2 Recruiter

Recruiters use TrueMatch to see past keyword noise — to find capable candidates the ATS would reject and to audit their own job descriptions.

Capability	What it does
Three-signal candidate view	Dual score (traditional vs capability) plus delta, with the capability narrative and supporting evidence per candidate.
Counter-recommendations	Hidden-gem candidates surfaced explicitly, with the reasoning for why capability exceeds keyword rank.
Credential-substitution panel	See which alternate evidence satisfies each role requirement, rated HIGH/MEDIUM/WEAK.
JD quality & interrogation	Score and fix job descriptions; flag inflated or contradictory requirements before posting.
JD simulation & evolution	Simulate how JD changes affect matching; track version drift over time.
Decision recording	Capture advance/decline decisions with rationale into an auditable trail.
Interview & scheduling	Two-way calendar integration with auto-scheduling and interview scorecards.
External ATS connectors	Bring candidates in from external ATS sources.
Recruiter metrics & DEI analytics	Pipeline metrics and diversity analytics over the recruiting funnel.

8.3 Administrator

Administrators operate, monitor and govern the platform.

Capability	What it does
User management	Manage accounts and roles across the organisation.
Billing console	Live Founding-100 inventory, revenue summary, order/fulfilment queue with status transitions, and refunds.
Governance reviews	Inspect assessments routed to human review by the governance gates.
Audit trail	Full, provenanced event history for compliance and legal defensibility.
Analytics & system metrics	Platform-wide analytics and operational health metrics.
Agent control	Configure and supervise autonomous ingestion and decision agents (24/7 operation).
Compliance tooling	Oversee PDPA/GDPR data-subject requests and encryption posture.

9. The Companion iOS App

The iOS app is a **native SwiftUI client** (iOS 17+, MVVM) that talks to the same backend over an authenticated API, with tokens held in the device keychain. It is **role-aware**: the tab bar adapts to whether the signed-in user is a candidate, recruiter or administrator.

The candidate experience is the most fully realised on mobile:

- **Assess** — upload or photograph a résumé and run an assessment (vision transcription lets a user snap a printed CV).
- **AI Assistant** — streaming conversational guidance.
- **CV Insights** — the career gap analysis.
- **Capability Translation** — the grounded ATS rewrite with before→after lift, evidence-grounded bullets and the honest "what we did not add" section, mirroring the web feature.
- **Jobs, Assessment History, Photo Resume Upload, Profile, Settings** — in the primary tabs and the "More" hub.

Recruiter and administrator tab sets expose their respective tools (JD simulation, decisions, agent control; users, audit, analytics). The app shares the platform's design system (cards, score gauges, delta bars, governance badges) for visual consistency, and supports deep links into specific results.

10. Security, Privacy & Data Governance

- **Encryption at rest** — résumé text, narratives, assessment internals, the capability-translation output and the pasted job descriptions are encrypted in the database; sensitive identifiers use a keyed blind index for lookup without exposure.
- **Authentication** — token-based sessions with refresh; Singapore **Singpass** login supported, storing only an opaque identifier — never the NRIC.
- **Least-data principle** — shareable results carry no personal data; governance internals never leak to clients.
- **Data-subject rights** — built-in export (portability) and erasure with cascade.
- **Regulatory alignment** — controls mapped to PDPA, GDPR, the EU AI Act and NYC Local Law 144.

11. Reliability & LLM Redundancy

All AI reasoning funnels through a single client layer with **bounded retries, a circuit breaker, and centralised cost/latency accounting**. To remove single-vendor dependency, the platform runs a **dual-LLM backbone**: a primary provider with **automatic failover** to a secondary provider when the primary fails (credit exhaustion, outage, or an open circuit).

Failover covers every reasoning path — structured assessment output, plain text, document-image transcription (vision) and streaming chat — while preserving the structured-output guarantee through forced tool use. Streaming fails over only **before** the first token is delivered, so a user never sees duplicated output. The provider used is recorded with each result, so cross-provider runs remain labelled and auditable.

Note on scoring variance: the deterministic keyword and semantic signals are identical regardless of provider; the AI-reasoned capability signal can vary between providers, so results are compared within, not across, providers.

12. Current Status

The platform is feature-complete for its launch scope and running in a local staging configuration for beta and the Founding-100 programme. The web build is green across all role areas; the iOS app builds and runs; the backend test suite and reasoning evaluations pass. Payments are wired and verified end-to-end in test mode. The dual-LLM failover is live and verified. The **external-evidence verification layer** (Appendix B) is implemented, regression-tested and validated live on real publication and patent records. Outstanding go-live items are operational/commercial (production hosting, data-residency confirmation for any secondary AI provider, and legal/entity setup) rather than engineering.

Appendix A — Live Validation: A Worked Example

To validate the platform on real-world data, an anonymised **AI-executive candidate profile** (25 years APAC venture-building + hands-on AI architecture, patents and published research) was assessed against **six real, currently-advertised AI-leadership job descriptions** sourced from public job boards. The exercise demonstrates the three core mechanics in action.

Step 1 — Fast fit-rank (deterministic, sub-second, zero AI cost)

All six roles scored on the keyword (ATS) and semantic signals — instantly and reproducibly, with no LLM calls:

Role (company anonymised)	Keyword / ATS	Semantic	Blended
Director, AI & Enterprise Architecture	31	83	57
Director, Enterprise AI Enablement	21	91	56
Head of AI Visibility (SEO-centric)	21	79	50
Senior AI Transformation & Operations	15	78	47
Chief AI Officer (agency practice)	15	71	43
AI Transformation Owner	18	65	42

The signal in the numbers: keyword/ATS scores of **15–31** mean a conventional ATS would filter this candidate out of *every* role — while semantic scores of **65–91** reveal strong real fit. That gap is the hidden-gem signal the platform exists to surface.

Step 2 — Deep-dive (capability translation on the top 3)

The top-ranked roles were run through the full capability-translation pipeline, which re-expresses the candidate's evidenced capability in each role's language and measures the before→after legibility lift:

Role	Keyword lift	Semantic lift	Verdict
Director, Enterprise AI Enablement	21 → 39 (+18)	91 → 94	Strongest fit
Director, AI & Enterprise Architecture	31 → 35 (+4)	83 → 88	Strong; gated by formal EA credential
Head of AI Visibility	21 → 33 (+12)	79 → 82	Demoted — critical domain gap

Step 2b — The three-signal verdict (top role)

For the strongest match, the full three-signal assessment was run. The signals deliberately tell **three different truths** — which is why the platform never collapses them into one number:

Signal	Score	What it means
Keyword / ATS	21 → 39	What an automated filter credits — translation makes the résumé legible
Semantic overlap	91 → 94	How strongly the experience maps to the role conceptually
Capability (verdict)	62	The rigorous, evidence-grounded judgment of real fit — never inflated
Delta (capability – ATS)	+41	The hidden-gem gap: capability the automated filter cannot see

Note the honesty in the data: the capability verdict (**62**) sits below the semantic overlap (91). The candidate's experience maps strongly, but the considered judgment of fit for this specific mandate is measured — real, unusual depth alongside genuine gaps. A single blended "match %" would erase exactly this nuance.

Step 3 — The honesty discipline (the differentiator)

Zero fabricated statements were produced across all three rewrites. The system surfaced genuine, non-negotiable gaps rather than inventing credentials:

- **Director, Enterprise AI Enablement** — verified strong alignment (25+ yrs vs the 10+ required; 4+ yrs as Chief AI Architect vs the 3–5 required), flagging only specific tooling familiarity as partial evidence.
- **Director, AI & Enterprise Architecture** — explicitly flagged the absence of a TOGAF / formal enterprise-architecture credential, and rated the AI-governance claim WEAK for lack of verified committee-leadership evidence.
- **Head of AI Visibility** — although this SEO-centric role ranked third on raw semantics, the capability layer demoted it, stating plainly there is "no evidence of SEO experience." A naïve semantic match was correctly caught as a false-positive.

This is the platform's promise demonstrated end-to-end: **surface the capability an ATS hides, translate it into legible language, measure the lift, and stay honest about what is genuinely absent** — the combination that separates legitimate capability translation from keyword-stuffing.

Engineering note. This live exercise also surfaced and resolved a latent reliability issue in the structured-output layer (large, content-dense résumés could truncate a parse and yield an empty result). The fix — a retry-on-truncation guard with regression coverage — hardens assessment and CV-analysis quality platform-wide, independent of which AI model is in use.

Appendix B — External-Evidence Verification & Trust Tiers

A capability verdict is only as trustworthy as the evidence under it. Most platforms treat everything on a résumé as equally true; TrueMatch does not. This appendix documents how the platform converts self-asserted claims into **graded, provenanced evidence** — and, just as importantly, how it **refuses to over-claim** when independent confirmation is not available.

B.1 The problem it solves

In the Appendix A worked example, several strong claims were rated **MEDIUM** rather than HIGH. Investigation confirmed this was **correct behaviour, not a weakness**: the claims were genuine but carried **no independent corroboration** in the assessment input. The lesson shaped a clear principle — evidence strength must reflect **who** attests to a claim, not merely how impressive it sounds. That principle is implemented as an explicit three-tier trust model.

B.2 The three-tier trust model

Trust tier	Source of attestation	Effect on evidence strength
Independently verified	An authoritative third-party registry confirms the claim (e.g. a DOI-registered, ORCID-confirmed publication; a published patent record; a GitHub repository).	May support a HIGH rating.
Candidate-corroborated	An official supporting document the candidate supplies confirms the claim (e.g. a patent-office acknowledgement letter), but the candidate controls the document.	May lift a claim to MEDIUM — never to HIGH on its own.
Unverified claim	The résumé asserts it; nothing external confirms it.	Treated as a claim; weighted accordingly.

The distinction between the first two tiers is the heart of the design. A document hosted or supplied by the candidate is **corroboration, not independent verification** — the candidate controls it — so it is held to a lower ceiling than a third-party registry. This is the same discipline that keeps the platform honest on the candidate side (Capability Translation never fabricates); here it keeps the **evidence** side honest.

B.3 Verification sources

Each source is an open, authoritative endpoint requiring no candidate cooperation beyond an identifier. Every fetch is independently guarded, so one unreachable source never fails the batch; all of it is a safe no-op when enrichment is disabled.

Source	What it confirms	Tier
ORCID	A researcher's verified identity and registered works.	Verified

Source	What it confirms	Tier
Zenodo · DataCite · Crossref	Published / deposited works by DOI — title, date, type, authorship.	Verified
GitHub	Real repositories, activity and ownership.	Verified
Published patent record (WIPO / Lens.org)	A patent application once it publishes in the public record.	Verified
Official supporting documents	A patent-office acknowledgement (or similar) the candidate supplies — application numbers, issuing authority, dates.	Corroborated

B.4 Publication discovery by identity

A candidate rarely lists every DOI. The discovery engine therefore searches the open registries by the candidate's **ORCID and name** to find their published work automatically — verifying it even when their own profile does not aggregate it. To prevent the obvious failure mode (attributing a stranger's work to the candidate), discovery applies a **misattribution guard**: a result is accepted only when a single author entry matches both the surname **and a distinctive given name**, with ubiquitous name particles ignored.

This guard earned its place in live testing. A name search for the Appendix A profile returned over a dozen works that were **not the candidate's** — they belonged to different people who merely shared a common surname. The naïve approach would have credited all of them. The guard correctly rejected every one, leaving only the works confirmed via the candidate's verified ORCID identity. The platform's rule is explicit: **under-claim before mis-attribute**.

B.5 Corroborating non-API claims (e.g. patents)

Some claims cannot be checked against a machine API — a national patent register may expose no public endpoint, and a freshly filed application has not yet published. For these, the candidate can supply the **official supporting document** (e.g. the patent office's acknowledgement letter). The platform fetches it, confirms the issuing authority's markers are present, and **extracts the verifiable identifiers** (application numbers, dates) for a recruiter to check directly. The claim is recorded as **corroborated** (MEDIUM ceiling) and is set to **upgrade automatically to verified** once the application publishes in the public patent record.

Security by construction. A candidate-supplied document is untrusted input. The platform extracts only **structured identifiers** from it and never feeds its free text into the reasoning model — so a document containing adversarial instructions (a "prompt-injection" attempt to inflate a score) has no surface on which to act. Detection is content-based (it inspects the file's actual type, not its name or extension), so documents from any file host are handled robustly.

B.6 Live validation

The verification layer was exercised end-to-end on the same anonymised AI-executive profile from Appendix A, against two senior, currently-advertised AI-deployment roles:

- **Nine** of the candidate's publications were independently **verified** (DOI-registered, ORCID-confirmed) — one more than the candidate had listed.
- **Six** patent applications were **corroborated** from their official patent-office acknowledgement letters: issuing authority confirmed, all six application numbers extracted, set to auto-upgrade on publication.
- Over a dozen same-surname publications by **other people** were correctly **rejected** by the misattribution guard.

Re-running the assessment **with** this verified evidence produced an honest, asymmetric result — exactly as intended:

Role	Capability (before → after)	HIGH achievements	Why
AI Deployment Lead (consulting)	47 → 52	4 of 10	Verified research + corroborated IP lifted technical-depth claims from MEDIUM to HIGH.
AI Deployment Strategist (agentic platform)	62 → 62 (held)	2 of 11	Evidence confirmed AI credibility, but the limiter is a genuine consulting-pedigree / domain gap — which verification cannot, and should not, paper over.

The asymmetry is the proof point. Verification **raised** the score where the ceiling was missing evidence (the Lead role), and **left it unchanged** where the ceiling was a real capability gap (the Strategist role). A system that simply rewarded "more evidence" with "higher score" would have inflated both. TrueMatch raised one and held the other — because that is the truthful answer.

B.7 Why this matters commercially

- **Defensibility** — every HIGH rating traces to a named, independent authority, recorded in the audit trail. A hiring decision built on it is legally and reputationally defensible.
- **Trust** — recruiters can rely on a HIGH rating precisely because the platform is demonstrably willing to withhold it.
- **Differentiation** — graded, provenanced, identity-discovered evidence with an explicit anti-misattribution discipline is materially beyond résumé keyword-matching, and beyond what the surveyed market offers.

— End of Briefing —